

Figure 8(a): High-Confidence Validation Predictions (MobileNetV2 >99% Posterior Confidence)

TN: Cat
(True: Cat, Pred: Cat, $p=0.002$)



TN: Cat
(True: Cat, Pred: Cat, $p=0.005$)



TP: Dog
(True: Dog, Pred: Dog, $p=0.998$)



TP: Dog
(True: Dog, Pred: Dog, $p=0.996$)

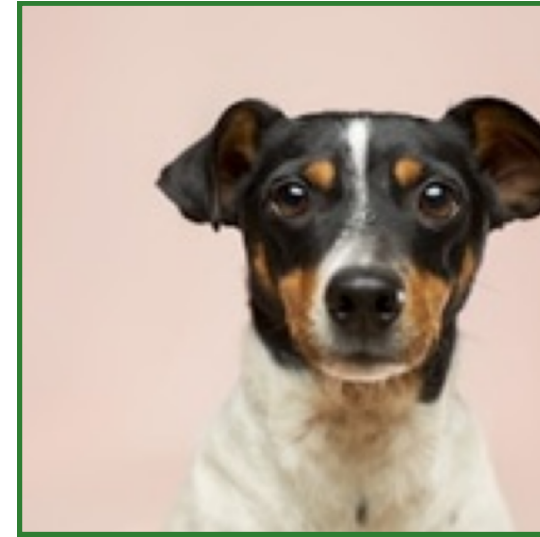


Figure 8(b): Failure Topology & Edge Cases (Validation Errors: 55 FPs, 59 FNs & Hard Invariant Cases)

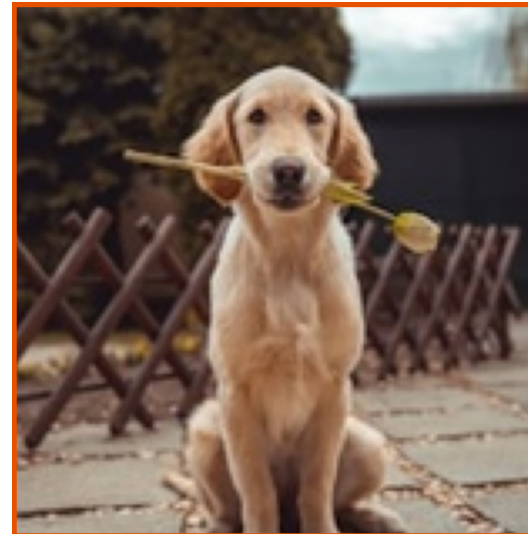
FP: False Dog
(True: Cat, Pred: Dog, $p=0.824$)



FN: False Cat
(True: Dog, Pred: Cat, $p=0.214$)



Ambiguous: Severe Occlusion
(True: Cat, Pred: Dog, $p=0.652$)



Ambiguous: Low Contrast
(True: Dog, Pred: Cat, $p=0.282$)

